Human memory → Smart

Computer mel. → Capacity limited

Memory – based on location

1) Internal → if process can directly access – RAM, ROM

2) External → can not → SSD, HDD

↓ I/O device

Peripherals

I/O module

Based on capacity

by bits / bytes / word

GB MB

Transfer

by bytes/words

by blocks*

Access method

magnetic tape

1) Sequential

2) Direct

3) Random → RAM

Access by address

4) Associative → memory content will be searched rather than address

↓ (cache) fast

Latency

1) Access time.

2) Memory cycle Address

3) Transfer time (T_B)

Ready to perform R/W

T_{A_1} Access time T_{A_2}

Total time = memory cycle time

$T_n = T_A + \frac{n}{R}$

Based on Materials

1) Semiconductor

2) Magnetic field

3) Optics

4) Magneto-optics

Volatile

Non-volatile

Erasable

Non-erasable

Registers

M.M

Bus

External Memory

Registers

cache

M.M

External Memory

Hierarchy

PCI → connection → can keep external memory (4 slot)

CPU

RA

Add

Cash

Data

System bus

m.m

RA

Then this

If not found, it's called cache miss. and then it will search M.M. Then M.M will update the cache.

Tag (contains address of M.M and control bit)

Cache

Line

line = K words

For each line, some extra information is present.

Address of M.M.

m line

M Block

$m < c M$

M.M

word

n bits

block → made of K words

$2^n - 1$

$\text{block} = \frac{2^n}{K}$ in M.M

Principal of locality